

An AI-Driven Personalized Curriculum Aligned Sinhala Teaching Assistant for Sri Lankan Secondary Education

Abstract—Sri Lanka faces a critical shortage of qualified teachers, particularly in rural regions, national shortage over 30,000 qualified teachers and student-to-teacher ratios that frequently exceed 50:1, resulting in profoundly unequal learning opportunities for General Certificate of Education Ordinary Level (GCE O/L) and Advanced Level (GCE A/L) students. Existing digital platforms provide static, prerecorded content and lack adaptive personalization, while general purpose Large Language Models (LLMs) lack Sri Lankan curriculum alignment and cannot dynamically assess or respond to individual learner states. To bridge this gap, this final year project introduces an Artificial Intelligence (AI) driven Sinhala teaching assistant that operates natively in the Sinhala language, strictly adheres to the national syllabus, dynamically adapts to each student's evolving knowledge state, and continuously monitors real-time learner engagement through advanced computer vision techniques.

This system architecture integrates seven core components: (i) a curriculum aligned Sinhala LLM (SinLlama-8B) for educational content generation, (ii) a Hybrid Bayesian Knowledge Tracing (Hybrid-BKT) personalization engine combining Personalized Clustered Bayesian Knowledge Tracing (PC-BKT) with Long Short-Term Memory (LSTM) based temporal modeling to adaptively estimate student mastery level, (iii) a Retrieval-Augmented Generation (RAG) pipeline that grounds the LLM in verified educational material using ChromaDB, (iv) a LangGraph orchestrated multi agent architecture coordinating eight specialized agents. Orchestrator, Evaluator, Personalization, Content Generator, Quiz generator, Explainer, Progress Tracker, and Synchronization agent, (v) a dedicated Student Engagement Detection Engine provides real-time, computer vision-based monitoring of learner attentiveness. (vi) a 3D avatar teacher for delivers lessons in a same human like manner with lip synchronization and voice variations, and (vii) an administrative portal for lesson management with automated text extraction pipeline, a mobile app for parents to monitor students' performance. Together, these components constitute a production-grade intelligent tutoring system designed to deliver personalized Sinhala medium education to students across Sri Lanka.

First, a curriculum aligned Sinhala LLM (SinLlama-8B) serves as the foundational educational content generator. Prior to its selection, five Sinhala supporting LLMs were fine-tuned using Low-Rank Adaptation (LoRA) on a custom dataset derived from the national curriculum and rigorously evaluated. SinLlama-8B achieved a ROUGE-1 score of 0.7714, a BERTScore of 0.9424, and an LLM-as-a-Judge score of 15.88/20, significantly outperforming alternative models such as Qwen-14B, Llama-3.1-8B, and DeepSeek-14B, and Mistral-7B. SinLlama-8B model was further fine-tuned on instruction following data to enhance its ability to

generate pedagogically structured Sinhala explanations that closely replicate the clarity and coherence of human teaching.

The System features a Hybrid Bayesian Knowledge Tracing (Hybrid-BKT) personalization engine that adaptively estimates student mastery level. This engine employs a weighted ensemble combining PC-BKT and LSTM-BKT. Hybrid-BKT enables the system to leverage K-Means++ clustering students into mastery levels for personalization, while capturing complex temporal learning dynamics through a two-layer stacked LSTM.

The Retrieval Augmented Generation (RAG) pipeline grounds the LLM in verified educational material. National textbooks were digitized, chunked into 512-token segments with a 64 token overlap, and embedded using the multilingual-e5-base model into a ChromaDB vector store. This pipeline employs document order retrieval to preserve lesson coherence and vector-similarity retrieval for dynamic quiz generation and lesson content generation. The production system uses a dual-LLM architecture: fine-tuned SinLlama-8B model handles curriculum grounded lesson content generation through RAG, while Llama-3.3-70B powers rapid agent orchestration.

LangGraph orchestrated multi agent architecture coordinates eight specialized agents. The Orchestrator, which serves as the entry point and conditional router, the evaluator, responsible for grading and mastery classification, the personalization agent, which tailors content to individual learner profiles. The content generator agent, which produces RAG grounded lesson material. Quiz agent, dynamically generates assessments. The Explain Agent which simplifies concepts for avatar delivery, the Progress Tracker Agent, which logs and monitors and the Speech synchronization agent, which manages whisper based forced alignment for Avatar speech synchronization. Together, these agents govern a four-phase adaptive teaching lifecycle: (i) Pre-assessment, (ii) Adaptive Content Generation, (iii) post-assessment, and (iv) Continuous Adaptation.

An interactive 3D avatar teacher, rendered via Three.js and streamed through WebRTC, delivers spoken Sinhala explanations synthesized using Google Gemini TTS, with Whisper-based forced alignment enabling real-time lip synchronization and sentence highlighting during avatar speech. A dedicated Student Engagement Detection Engine operates as an independent computer vision microservice, utilizing MediaPipe, a custom CNN for emotion recognition, Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR), head pose estimation, and YOLOv8-based phone detection to classify student engagement as Highly Engaged, Moderately Engaged, Low Engagement, or Not Engaged. The full system is accessible via two Next.js web applications and a mobile application. One web application for students and teachers, an administrative portal for system management and lesson

content ingestion through an automated textbook digitization pipeline, in which national curriculum textbooks are extracted, chunked, and indexed into the RAG vector store. A React Native mobile application further enables parents to monitor their child's learning progress, mastery levels, and engagement history in real time.